Module 04: Cognition — Deep Temporal Models and Hierarchical Inference

> **Course**: Active Inference 401 | **Unit**: Advanced Theory | **Audience**: Graduate students / researchers

Learning Objectives

1. Analyze **deep temporal models** — hierarchical generative models with multiple temporal scales.
2. Evaluate the mathematical structure of **hierarchical message passing** — ascending and descending messages across levels.
3. Examine the role of **temporal depth** in planning, imagination, and counterfactual reasoning.

Key Concepts

1. The Problem of Temporal Depth

Simple generative models update beliefs moment-by-moment. But intelligent behavior requires modeling the past and future over long horizons:

Temporal abstraction: The world operates at multiple timescales simultaneously — from millisecond neural events to decadal life plans. A flat model treating all time points equally is computationally intractable. Deep temporal models address this through hierarchy.

The key architecture: A deep temporal model has L levels, each operating at a progressively slower timescale:

- Level 1: Fast dynamics (sensory fluctuations, milliseconds)
- Level 2: Medium dynamics (events, seconds to minutes)
- Level 3: Slow dynamics (contexts, hours to days)
- Level L: Slowest dynamics (life narratives, years)

Each level generates predictions about the level below, and receives prediction errors from it.

2. Hierarchical Generative Model Structure

The mathematical structure of a deep temporal model:

State at level l, time t: $s_t^{(l)}$

Transition model: $s_{t+1}^{(l)} \sim p(s_{t+1}^{(l)} \mid s_t^{(l)}, s_t^{(l+1)})$

States at level l transition based on their own dynamics AND the contextual influence of the level above.

Observation model: Only the lowest level generates observations:

$o_t \sim p(o_t \mid s_t^{(1)})$

Key property: Higher levels evolve more slowly because they represent more abstract, temporally stable features. The "clock speed" decreases with level — higher levels change less frequently.

3. Hierarchical Message Passing

Inference in deep temporal models involves message passing between levels:

Ascending messages (bottom-up): Prediction errors from level l to level l+1. These carry information about what the lower level's predictions got wrong, compressed into summary statistics that inform the higher level's context estimation.

Descending messages (top-down): Predictions from level l+1 to level l. These carry the higher level's expectations about what should be happening at the lower level, constraining lower-level inference.

Update equations: At each level l, the belief update integrates:

- Prediction error from below (ascending): $\varepsilon^{(l-1)} = o - g(s^{(l)})$ or $s^{(l-1)} - f(s^{(l)})$
- Contextual prior from above (descending): $p(s^{(l)} \mid s^{(l+1)})$
- Lateral dynamics: $p(s_t^{(l)} \mid s_{t-1}^{(l)})$

The precision weighting determines how much weight each message receives.

4. Planning as Inference in Deep Temporal Models

Planning in Active Inference is formalized as inference about future actions (policies). Deep temporal models extend this:

Multi-scale planning:

- Level 1: "Should I move my hand left or right" (motor-level policy)
- Level 2: "Should I reach for the cup or the phone" (action-level policy)
- Level 3: "Should I have coffee or tea" (goal-level policy)
- Level L: "Should I pursue career A or career B" (life-level policy)

Higher-level policies constrain lower-level policies through descending messages.

Expected free energy at multiple scales: Each level evaluates its own expected free energy:

$G^{(l)}(\pi) = E_q[\ln q(s^{(l)}_\tau \mid \pi) - \ln p(o_\tau, s^{(l)}_\tau \mid \pi)]$

Higher levels consider outcomes over longer horizons, using more abstract representations.

5. Imagination and Counterfactual Reasoning

Deep temporal models enable imagination — the generative model can be "run forward" without sensory input:

Generative replay: The model generates predicted future trajectories by sampling from the transition model at each level. This is mental simulation — evaluating what would happen under different policies.

Counterfactual reasoning: By "clamping" different policy choices and running the model forward, the agent can evaluate what *would* happen if it did X vs. Y. This corresponds to the expected free energy calculation — comparing $G(\pi_1)$ vs. $G(\pi_2)$.

Dreaming and creativity: During sleep or rest, the model can be run in "generative mode" without constraining observations. This produces novel combinations of states — the basis for creativity and insight.

Summary

Deep temporal models address the challenge of multi-scale temporal inference through hierarchical generative models with ascending (bottom-up prediction error) and descending (top-down prediction) messages. Planning at multiple scales is implemented as inference about nested policies. Imagination and counterfactual reasoning emerge from running the generative model forward without sensory constraint.

Further Reading

- Friston, K. et al. (2017). Deep temporal models and active inference. *Neuroscience & Biobehavioral Reviews*, 77, 388-402.
- Hesp, C. et al. (2021). Deeply felt affect. *Psychological Review*, 128(4), 723-760.
- Parr, T. & Friston, K. (2018). The anatomy of inference: Generative models and brain structure. *Frontiers in Computational Neuroscience*, 12, 90.